



# Stripe Payment Integration Testing Guide

## Quick Access

- **Live Site:** <https://rrg-solutions.com>
- **Admin Login:** <https://rrg-solutions.com/admin/login>
- Email: `admin@energenius.com`
- Password: `admin123`
- **Stripe Dashboard:** <https://dashboard.stripe.com/test>



## Pre-Testing Checklist



### Webhook Configuration (CRITICAL)

1. **Navigate to:** <https://dashboard.stripe.com/test/webhooks>
2. **Add endpoint:** <https://rrg-solutions.com/api/webhooks/stripe>
3. **Select events:**
  - `checkout.session.completed`
  - `payment_intent.succeeded`
  - `payment_intent.payment_failed`
4. **Verify signing secret matches:**  
`STRIPE_WEBHOOK_SECRET=whsec_P0t2Be7lh2RKxaF0B7IpUTVcVw4xhKL6`



## Test Card Numbers



### Successful Payments

| Card Number                      | Description          | Use Case                    |
|----------------------------------|----------------------|-----------------------------|
| <code>4242 4242 4242 4242</code> | Visa (Success)       | Standard successful payment |
| <code>5555 5555 5555 4444</code> | Mastercard (Success) | Alternative success test    |
| <code>3782 822463 10005</code>   | Amex (Success)       | 4-digit CVC test            |

## ✖ Declined Payments

| Card Number         | Error Type         | Use Case                |
|---------------------|--------------------|-------------------------|
| 4000 0000 0000 0002 | Generic decline    | Test decline handling   |
| 4000 0000 0000 9995 | Insufficient funds | Test insufficient funds |
| 4000 0000 0000 9987 | Lost card          | Test lost/stolen card   |
| 4000 0000 0000 9979 | Stolen card        | Test fraud prevention   |
| 4000 0000 0000 0069 | Expired card       | Test expiry validation  |

## ⚠ Special Scenarios

| Card Number         | Behavior                | Use Case               |
|---------------------|-------------------------|------------------------|
| 4000 0025 0000 3155 | Requires authentication | Test 3D Secure         |
| 4000 0000 0000 0341 | Attach/charge failure   | Test processing errors |

**Note:** Use any future expiry date (e.g., 12/25 ), any 3-digit CVC (e.g., 123 ), and any ZIP code (e.g., 12345 )

## 🎯 Testing Scenarios

### Scenario 1: Full Payment - Success ✓

**Objective:** Test complete invoice payment

#### 1. Create Invoice:

- Go to Admin Dashboard → Invoices → New Invoice
- Customer: "John Doe"
- Email: "john@test.com"
- Add Product: "EnerGenius Guardian 5000" (\$4,899)
- Tax: \$367.43 (7.5%)
- **Total:** \$5,266.43
- Save Invoice

#### 2. Make Payment:

- Click "Pay Full Amount" button
- Use card: 4242 4242 4242 4242
- Expiry: 12/25 , CVC: 123 , ZIP: 12345
- Click "Pay"

#### 3. Expected Results:

- ✓ Redirect to success page
- ✓ Toast notification: "Payment successful!"
- ✓ Invoice status: "Paid"

- ☒ Payment recorded in database
  - ☒ Stripe dashboard shows successful charge
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## Scenario 2: Deposit + Balance ☒

**Objective:** Test split payment (deposit first, balance later)

### Step 2A: Pay Deposit

1. **Create Invoice:**

- Total: \$10,000
- Deposit: \$3,000 (30%)
- Balance: \$7,000

2. **Pay Deposit:**

- Click "Pay Deposit (\$3,000)" button
- Use card: 4242 4242 4242 4242
- Complete payment

3. **Expected Results:**

- ☒ Status: "Partially Paid"
- ☒ Deposit Paid: \$3,000
- ☒ Balance Due: \$7,000
- ☒ "Pay Balance" button appears

### Step 2B: Pay Balance

1. **Pay Balance:**

- Click "Pay Balance (\$7,000)" button
- Use card: 5555 5555 5555 4444
- Complete payment

2. **Expected Results:**

- ☒ Status: "Paid"
  - ☒ Total Paid: \$10,000
  - ☒ All payment buttons disabled
- 

## Scenario 3: Payment Declined ☒

**Objective:** Test error handling for declined cards

1. **Create Invoice:**

- Any amount (e.g., \$2,000)

2. **Attempt Payment:**

- Click "Pay Full Amount"
- Use **declined card:** 4000 0000 0000 0002
- Try to complete payment

3. **Expected Results:**

- ☒ Payment fails at Stripe
- ☒ Error message: "Your card was declined"

- ❌ Invoice status remains "Pending"
- ❌ No payment recorded

## Scenario 4: Insufficient Funds ❌

**Objective:** Test specific decline reason

### 1. Create Invoice:

- Amount: \$5,000

### 2. Attempt Payment:

- Use card: 4000 0000 0000 9995
- Try to complete payment

### 3. Expected Results:

- ❌ Error: "Insufficient funds"
- ❌ Invoice remains "Pending"
- ❌ Webhook receives `payment_intent.payment_failed` event

## Scenario 5: Multiple Payment Attempts ↺

**Objective:** Test retry logic and status tracking

### 1. Create Invoice: \$3,000

### 2. First Attempt (Fail):

- Use declined card: 4000 0000 0000 0002
- Payment fails

### 3. Second Attempt (Success):

- Click payment button again
- Use success card: 4242 4242 4242 4242
- Payment succeeds

### 4. Expected Results:

- ✅ Invoice status updates to "Paid"
- ✅ Only successful payment recorded
- ✅ Failed attempts not counted

## Verification Checklist

After each test, verify:

### In EnerGenius Admin Dashboard:

- [ ] Invoice status updated correctly
- [ ] Payment amount recorded
- [ ] Deposit/Balance tracking accurate
- [ ] Payment method saved

- [ ] Timestamps correct

### In Stripe Dashboard:

- [ ] Payment Intent created
- [ ] Charge successful (or failed as expected)
- [ ] Webhook delivered successfully
- [ ] Customer email recorded
- [ ] Metadata includes invoice ID

### In Database (via Prisma Studio):

```
cd /home/ubuntu/energenius_website/nextjs_space
yarn prisma studio
```

- [ ] Invoice record updated
- [ ] Payment record created
- [ ] Relationships correct (Invoice ↔ Payment)



## Troubleshooting

### Issue: Webhook not firing

#### Solution:

1. Check webhook endpoint is active in Stripe Dashboard
2. Verify URL: `https://rrg-solutions.com/api/webhooks/stripe`
3. Test webhook manually in Stripe Dashboard
4. Check server logs for errors

### Issue: Payment succeeds but invoice not updating

#### Possible causes:

- Webhook signing secret mismatch
- Webhook events not selected correctly
- Database connection issue

#### Debug steps:

```
# Check webhook events in Stripe Dashboard
# Look for failed webhook attempts
# Verify signing secret matches .env
```

### Issue: “Payment processing” stuck

#### Solution:

- Check Stripe Dashboard for payment status
- Verify webhook was delivered
- Manually trigger webhook retry in Stripe Dashboard
- Check invoice status in database



## Expected Payment Flow

1. User clicks "Pay" button  
↓
2. API creates Stripe Checkout Session  
↓
3. User redirected to Stripe Checkout  
↓
4. User enters card details  
↓
5. Stripe processes payment  
↓
6. Webhook receives "checkout.session.completed"  
↓
7. API updates invoice status  
↓
8. User redirected back to invoice page  
↓
9. Success toast notification shown



## Key Metrics to Monitor

### During Testing:

- Payment success rate: Should be 100% with valid test cards
- Webhook delivery time: Typically < 3 seconds
- Invoice update latency: < 5 seconds after webhook
- Error handling: Graceful failures with clear messages

### Production Readiness:

- [ ] All test scenarios passed
- [ ] Webhooks configured and tested
- [ ] Error messages user-friendly
- [ ] Payment confirmations working
- [ ] Database updates atomic and consistent
- [ ] Ready to switch to live keys



## Going Live

When ready for production:

### 1. Get Live API Keys:

- Navigate to: <https://dashboard.stripe.com/apikeys>
- Copy Live Secret key (starts with sk\_live\_)
- Copy Live Publishable key (starts with pk\_live\_)

### 2. Configure Live Webhook:

- Go to: <https://dashboard.stripe.com/webhooks>
- Add endpoint: <https://rrg-solutions.com/api/webhooks/stripe>

- Select same events as test
- Copy Live `Signing secret` (starts with `whsec_`)

### 3. Update Environment Variables:

```
bash
```

```
STRIPE_SECRET_KEY=sk_live_...
```

```
NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY=pk_live_...
```

```
STRIPE_WEBHOOK_SECRET=whsec_...
```

### 4. Deploy:

- Test with real card first (small amount)
- Monitor Stripe Dashboard
- Verify webhooks firing

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## Support

### Stripe Documentation:

- Testing: <https://stripe.com/docs/testing>
- Webhooks: <https://stripe.com/docs/webhooks>
- Checkout: <https://stripe.com/docs/payments/checkout>

### EnerGenius Admin:

- Email: [admin@energenius.com](mailto:admin@energenius.com)
- Dashboard: <https://rrg-solutions.com/admin>

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**Last Updated:** December 24, 2024

**Integration Status:**  Complete - Ready for Testing